Drop Impact on a Solid Surface: VOF Simulation and Comparison with Gunjal *et al.* (2005)

AM6513 Multiphase Flow – Assignment 8
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1 Problem

A water drop falls onto a flat Teflon surface. After it touches the surface it spreads, pulls back (recoils), jumps up (rebounds), oscillates and finally sits still as a sessile drop. The task is to model this with a CFD solver and compare with the experiments and simulations of Gunjal, Ranade and Chaudhari [1], Figure 9. The two numbers reported there are:

- average oscillation period of the drop height: **16.57 ms**,
- final (equilibrium) drop height: **2.2 mm**.

Which case is Figure 9? The caption of Figure 9 says "Case 3". The text of the paper uses a separate numbering in which "Case 3" is the 0.3 m/s drop on Teflon. In Table 2 of the paper the same drop is listed as *Case 4* (water on Teflon, 110°, regime "Rebound"), while Table 2's Case 3 is water on glass, which only oscillates (11.33 ms, 1.6 mm, Fig. 7b). Figure 8 (Case 4) gives the same 16.57 ms and 2.2 mm, the text states that the rebound of this Teflon drop is shown in Figure 9, and the Figure 9 images show the drop leaving the surface. The inputs used here are therefore those of **Table 2, Case 4**.

Contact angle. Table 2 of the paper uses a static contact angle of 110° for Case 4, and this value is used for the main simulation. The paper also reports measured water-Teflon contact angles of 110–125° and 110–120°. To check how sensitive the results are to this value, the case was also run with 120°.

The flow is two-phase (water and air), incompressible, isothermal and laminar. Surface tension, gravity and wetting of the wall control the motion.

Code

`https://github.com/ae25d406-glitch/AM6513-MultiPhase.git`.

2 Domain and inputs

The drop is round seen from above, so the flow is axisymmetric. Only a thin 5° wedge around the vertical axis is solved (Fig. 1), which is the same 2D axisymmetric simplification the paper used for impacts below 1 m/s.

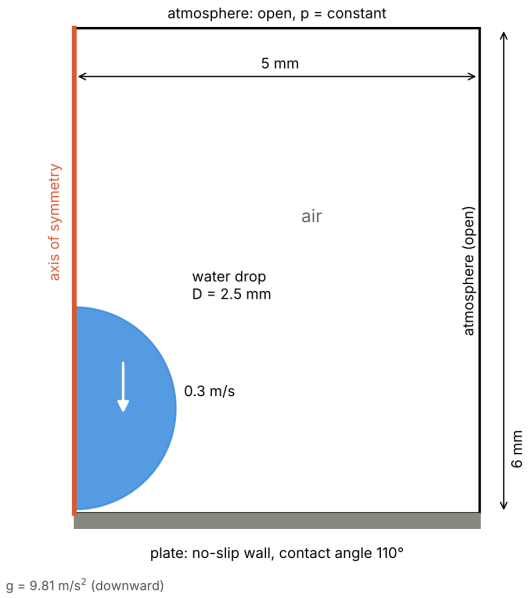


Figure 1: Computational domain (right half of the axisymmetric problem) and boundary conditions.

Table 1: Input data (Table 2, Case 4 of [1]; air properties are standard values, not given in the paper).

Quantity	Value	Source
Drop diameter D	2.5 mm	paper
Impact velocity V	0.3 m/s (downward)	paper
Water density, viscosity	1000 kg/m^3, 1×10^{-3} Pa s	paper
Surface tension σ	0.073 N/m	paper
Static contact angle θ (water–Teflon)	110° (main); 120° (sensitivity)	paper (Table 2; measured range 110–125°)
Air density, viscosity	1.2 kg/m^3, 1.8×10^{-5} Pa s	assumed
Gravity g	9.81 m/s^2	–
Re, We, Bo	750, 3.12, 0.85	paper
Domain	5 mm (radius) × 6 mm (height)	paper

Table 2: Boundary and initial conditions.

Boundary	Condition
Plate (bottom)	no-slip wall, $\mathbf{U} = 0$; contact angle 110° on α; zero flux for pressure
Atmosphere (top, side)	constant pressure; air may flow in or out
Front / back	wedge (axisymmetry)
Initial state	air at rest everywhere; cells inside a sphere of radius 1.25 mm set to water ($\alpha = 1$) with $\mathbf{U} = (0, -0.3, 0)$ m/s; bottom of the drop 0.05 mm above the plate

3 Methodology

3.1 Governing equations

One set of equations is solved for both fluids. The volume of fluid (VOF) [3] field α is the fraction of water in a cell ($\alpha = 1$ water, $\alpha = 0$ air, $0 < \alpha < 1$ interface). Density and viscosity are mixed with α:

$$\rho = \alpha\rho_w + (1-\alpha)\rho_a, \qquad \mu = \alpha\mu_w + (1-\alpha)\mu_a. \tag{1}$$

Continuity, momentum (Navier–Stokes) and the transport of α:

$$\nabla \cdot \mathbf{U} = 0, \tag{2}$$

$$\frac{\partial(\rho\mathbf{U})}{\partial t} + \nabla\cdot(\rho\mathbf{U}\mathbf{U}) = -\nabla p + \nabla\cdot\left[\mu\left(\nabla\mathbf{U} + \nabla\mathbf{U}^T\right)\right] + \rho\mathbf{g} + \mathbf{F}_\sigma, \tag{3}$$

$$\frac{\partial\alpha}{\partial t} + \nabla\cdot(\mathbf{U}\alpha) + \nabla\cdot\left[\mathbf{U}_r\,\alpha(1-\alpha)\right] = 0. \tag{4}$$

The last term in the α equation is a compression term that keeps the interface sharp. Surface tension is added with the continuum surface force (CSF) model of Brackbill *et al.* [2]:

$$\mathbf{F}_\sigma = \sigma\kappa\nabla\alpha, \qquad \kappa = -\nabla\cdot\left(\frac{\nabla\alpha}{|\nabla\alpha|}\right). \tag{5}$$

At the plate, the interface normal is tilted so that the water meets the wall at the contact angle θ; this is the wall-adhesion model (Eq. 11 of [1]). The solver works with $p_{rgh} = p - \rho\,\mathbf{g}\cdot\mathbf{h}$, the pressure without its hydrostatic part, so that pressure and gravity balance exactly in still fluid.

3.2 Numerical method

- Finite-volume method in OpenFOAM 13 [4]. Solver module: `incompressibleVoF`, run with `foamRun`.
- Time: first-order implicit (Euler). Momentum convection: second-order linear-upwind. Diffusion and gradients: second-order central.
- α equation: van Leer scheme with interface compression, solved with MULES (keeps $0 \le \alpha \le 1$).
- Pressure–velocity coupling: PISO (PIMPLE with one outer loop), three pressure corrections per time step.
- Time step: adaptive, Courant number ≤ 0.2 and $\Delta t \le 4\,\mu$s. The capillary-wave limit $\Delta t < \sqrt{\bar\rho\,\Delta x^3/(2\pi\sigma)} \approx 1.2\times10^{-5}$ s for $\Delta x = 0.05$ mm, so 4 μs is safe.
- Mesh: uniform square cells, 20 cells/mm (100×120 cells, $\Delta x = 0.05$ mm). Simulated time 150 ms, results saved every 0.5 ms.

3.3 Solver features used

- `blockMesh`: one structured block collapsed onto the axis to form the 5° wedge; `wedge` patches give axisymmetry. `checkMesh` to check mesh quality.
- `setFields` with a `sphere` zone: places the drop ($\alpha = 1$) and its impact velocity.
- `incompressibleVoF` module: two-phase VOF with MULES and interface compression, CSF surface tension (from σ in `phaseProperties`), laminar flow.

- Boundary conditions: `contactAngle` (wall wetting), `noSlip` and `fixedFluxPressure` on the plate; `prghTotalPressure`, `inletOutlet` and `pressureInletOutletVelocity` on the atmosphere.
- Adaptive time stepping (`maxCo`, `maxAlphaCo`, `maxDeltaT`).
- `decomposePar` / `mpirun` for parallel runs.
- A Python script (NumPy, Matplotlib) for drop height, period, final height and plots.

3.4 Paper versus present set-up

Table 3: Model used in the paper and present work.

Item	Gunjal *et al.* [1]	Present work
Software	Fluent 6.0, mesh in GAMBIT 2.0	OpenFOAM 13 (`incompressibleVoF`)
Interface method	VOF, geometric PLIC reconstruction	VOF, algebraic MULES with interface compression
Surface tension	CSF	CSF
Wall wetting	static contact angle, 110°	static contact angle, 110°
Flow model	laminar, incompressible, Newtonian	same
Domain	2D axisymmetric, 5 × 6 mm	5° wedge (axisymmetric), 5 × 6 mm
Mesh	20 cells/mm, uniform near the drop, coarser further away	20 cells/mm, uniform everywhere
Time step	4 μs, first-order implicit	adaptive, $\leq$ 4 μs, first-order implicit
Space discretisation	second order	second order
P–U coupling	PISO, body-force-weighted pressure	PISO, p_{rgh} formulation
Boundaries	symmetry axis, no-slip wall, constant pressure	wedge, no-slip wall, constant pressure
Initial condition	sphere patched with $\alpha = 1$ and impact velocity	same (`setFields`)

The main difference is the way the interface is kept sharp: PLIC rebuilds a straight line in each interface cell, while MULES adds a compression flux. MULES was used here because it is the standard interface method of the OpenFOAM VoF solver (chosen on the suggestion of the AI tool, see Section 6). Both methods conserve the liquid volume.

3.5 Runs and post-processing

Table 4: Simulations performed (all 0–150 ms).

Run	Mesh	Contact angle	Purpose
1 (main)	20 cells/mm (100 × 120)	110°	comparison with the paper
2	10 cells/mm (50 × 60)	110°	grid study
3	20 cells/mm	120°	upper end of the measured range

A Python script reads α from every saved time and finds the $\alpha = 0.5$ surface. From it the drop height $h(t)$ (top of the drop), the spread radius and the wetted radius on the plate are measured. The oscillation period is the mean time between successive height peaks after the drop is back on the plate ($t > 40$ ms). The equilibrium height is the mean of h over the last three periods.

4 Results

4.1 Comparison with the experiment and the simulation of the paper

Three results are compared: (i) the experiment of Gunjal *et al.*, (ii) their VOF simulation, and (iii) our OpenFOAM simulation. The first two were taken from Figure 8 of the paper by reading the values off the two curves (accurate to about 1%). All three are plotted together in Fig. 2, with height divided by 2.2 mm and time divided by 16.57 ms, as in the paper.

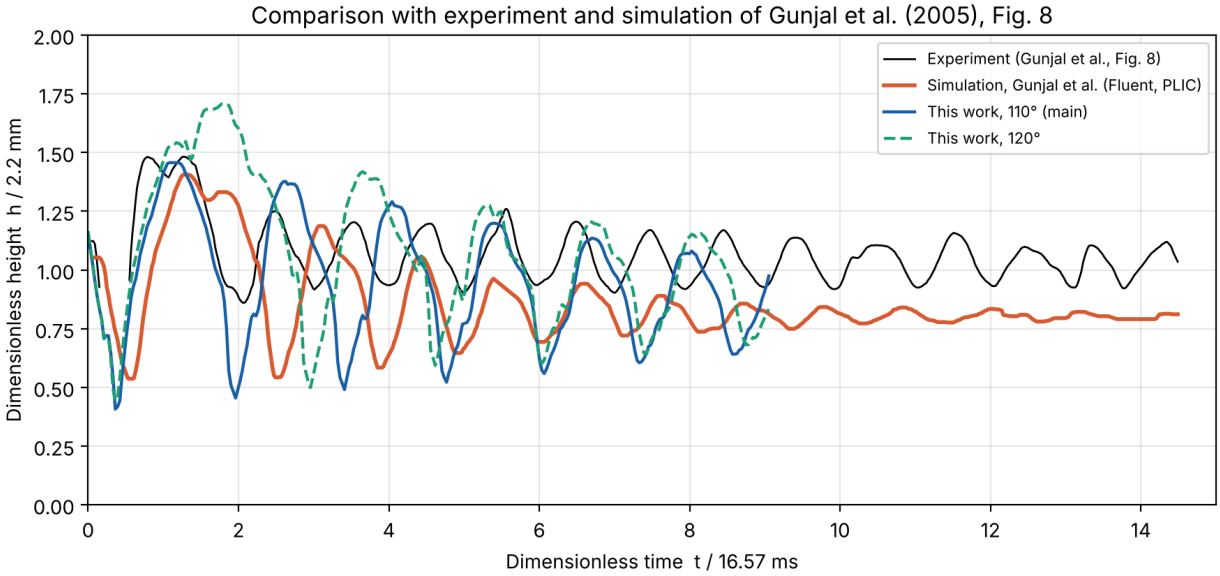


Figure 2: Drop height against time: experiment and simulation of Gunjal *et al.* [1] (digitised from their Fig. 8) and the present simulations. Our runs end at 150 ms ($t/16.57 \approx 9$).

Table 5: Comparison of the present results with the experiment and the simulation of the paper. Periods are taken over the same window (40–150 ms) for all curves. Values in brackets are differences from the experiment.

Quantity	Experiment [1]	Paper simulation [1] (Fluent, PLIC)	Present work, 110° (main)	Present work, 120°
Oscillation period (ms)	16.57	18.6 (+12%)	22.4 (+35%)	24.8 (+50%)
Final height (mm)	2.2	1.79 (−19%)	1.98 (−10%)	2.09 (−5%)
Max. height, 1st cycle (mm)	3.25	3.09 (−5%)	3.21 (−1%)	3.76 (+16%)
Max. spread diameter (mm)	3.9	not reported	3.69 (−5%)	3.55 (−9%)
Drop leaves the plate	yes	yes	no	yes (17.5–41 ms)

Main points:

- The paper's own simulation also has a longer period than the experiment (18.6 against 16.57 ms) and a lower final height (about 1.8 against 2.2 mm). Our model shows the same

two trends; the period error is larger (+35%), while the final height is closer to the experiment (−10% against −19%).

- The first-cycle maximum height of our main run (3.21 mm) is closer to the experiment than the paper's simulation (3.09 mm).
- In the experiment and in the paper's simulation the drop leaves the plate. Our main run does not; with 120° it does.

4.2 Drop shapes (Figure 9)

Figure 3 shows our drop at the same times as the simulation panels of Figure 9 in the paper; Table 6 compares the stages.

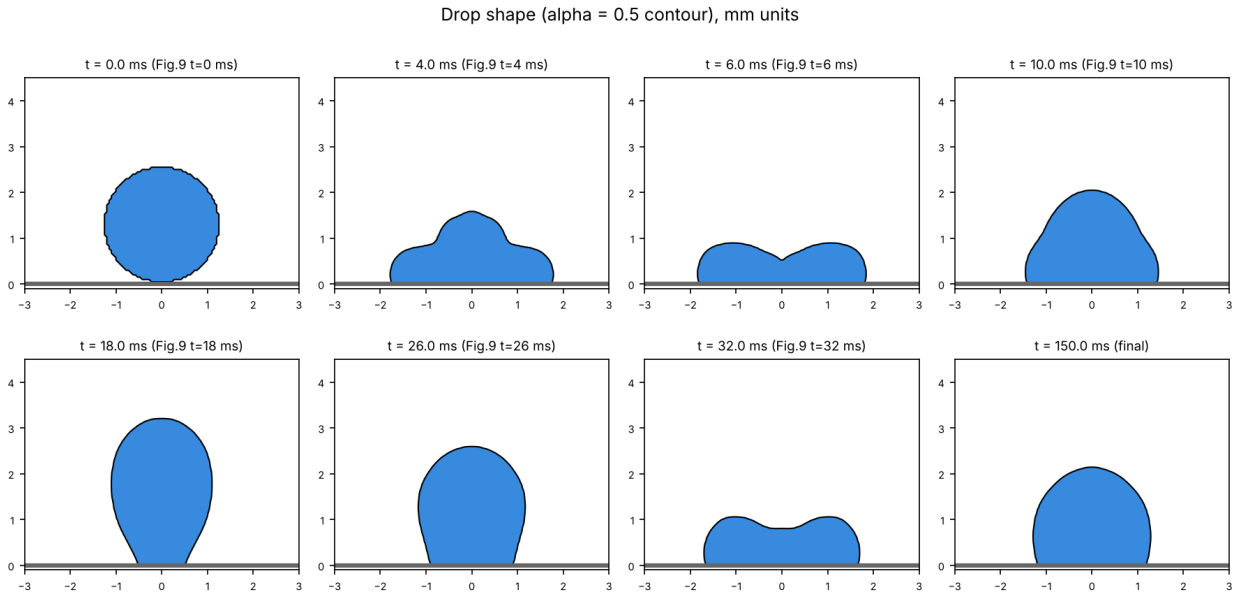


Figure 3: Run 1 (20 cells/mm, 110°): drop outline ($\alpha = 0.5$) at the times of Fig. 9 in [1].

Table 6: Stages of Figure 9 (paper) and the present simulation.

Time	Paper (simulation / experiment)	Present work, 110°	Present work, 120°
4 ms	flattened drop with a bump in the middle	same	same
6 ms	thin ring, dip at the centre	same, ring 3.7 mm wide	same
10 ms	recoil into a cone	same	same
18 ms	tall column, about to lift off (expt. 16 ms)	tall column 3.2 mm, base still on plate	column, just lifted off
26 ms	drop in the air (expt. 24 ms)	falling back, base on plate	drop in the air
32 ms	drop back on the plate, oscillating	flattened, oscillating	still in the air

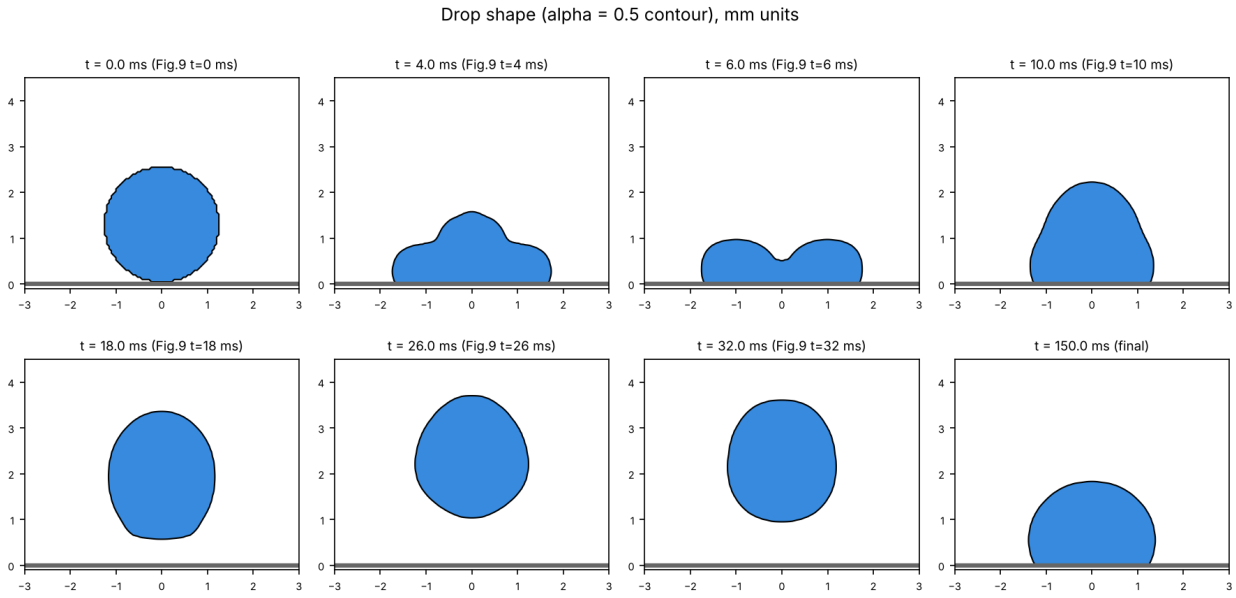


Figure 4: Run 3 (20 cells/mm, 120°): the drop leaves the plate between 17.5 and 41 ms, as in Fig. 9 of the paper.

The sequence of spreading, recoil and oscillation matches Figure 9. The main difference is the lift-off, which needs a slightly larger contact angle in our model. The paper's simulation panels also lag the photographs by about 2 ms (18/16, 26/24, 32/30 ms), which agrees with its longer period.

4.3 Grid study

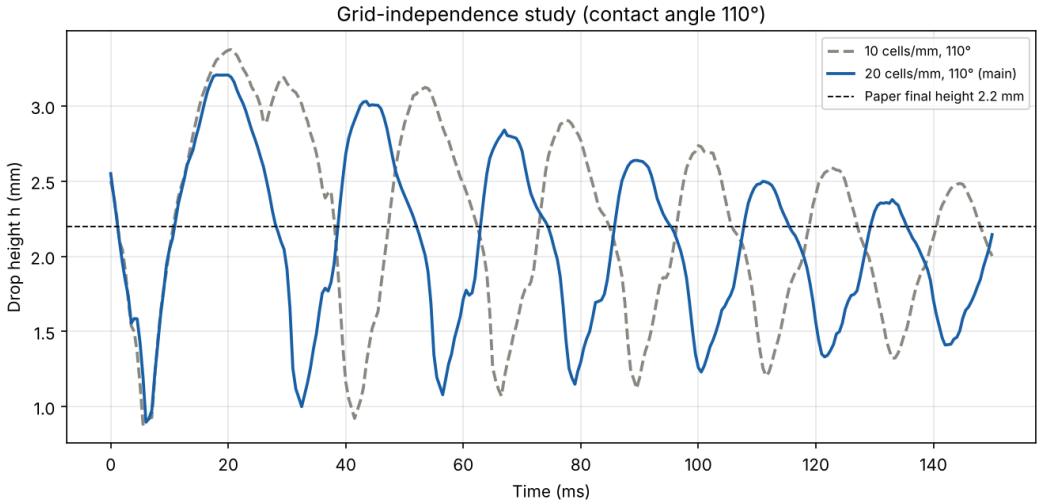


Figure 5: Drop height for 10 and 20 cells/mm (110°).

Table 7: Grid study (110°).

Mesh	Period (ms)	Final height (mm)	Max height (mm)	Max spread (mm)	Leaves plate
10 cells/mm	22.8	2.04	3.37	3.64	21.0–31.5 ms
20 cells/mm	22.4	1.98	3.21	3.69	no

Between 10 and 20 cells/mm the period changes by about 2% and the final height by about 3%, so these quantities are not sensitive to the mesh. Whether the drop briefly leaves the plate does depend on the mesh, because lift-off is decided by the thin neck of liquid at the contact line, which is only a few cells wide.

4.4 Effect of the contact angle

Table 2 of the paper gives 110°, but the paper also reports measured water–Teflon contact angles of 110–125° and 110–120°. The case was therefore repeated with 120°, keeping everything else the same.

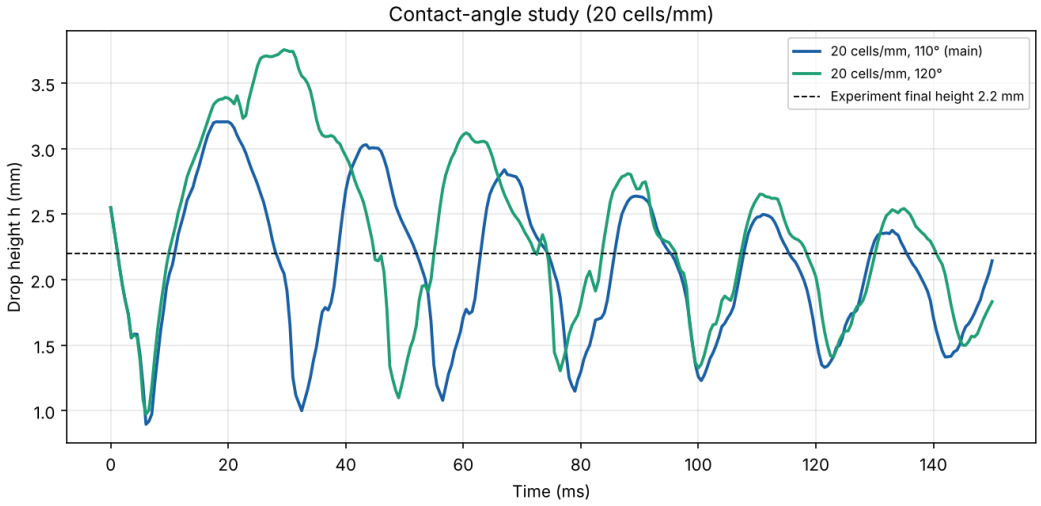


Figure 6: Drop height for contact angles of 110° and 120° (20 cells/mm).

Table 8: Contact-angle study (20 cells/mm). h_{cap} is the height of a resting spherical cap with the same volume at that angle.

Angle	Period (ms)	Final height (mm)	h_{cap} (mm)	Max height (mm)	Max spread (mm)	Leaves plate
110°	22.4	1.98	1.85	3.21	3.69	no
120°	24.8	2.09	1.98	3.76	3.55	once (17.5–41 ms)
Experiment [1]	16.57	2.2	–	3.25	3.9	yes

A larger contact angle makes the surface more water-repellent: at 120° the drop jumps higher, lifts off the plate (as in Figure 9) and settles taller, closer to 2.2 mm. However, its first-cycle height is now 16% too high and its period is longer. Neither angle matches every measurement: 110° gives the best first-cycle height and spread, while 120° gives the rebound and a final height closer to the experiment.

4.5 Mass conservation

The liquid volume changed by less than 0.07% over 150 ms in every run. On the 20 cells/mm mesh the starting volume is about 1% above that of an exact sphere because the drop is built from whole square cells.

5 Discussion

- **Static contact angle.** A real drop on Teflon has a larger angle when its edge advances and a smaller one when it recedes (the paper measured 110–125°). A single fixed angle cannot reproduce both, which explains why each angle in Table 8 matches some features but not others.
- **Final height.** A fully settled drop of this volume at 110° would be a spherical cap 1.85 mm tall, and gravity flattens it further. The simulated "final" heights are averages of a drop that is still oscillating at 150 ms and lie 0.1–0.15 mm above the cap value, so a longer run would lower them. A settled height of 2.2 mm points to an effective contact angle of about 140°, which is above the 110° given in Table 2 of the paper.
- **Period.** All runs give a longer period than the experiment. The paper's own simulation does the same (18.6 ms, Table 5); the authors state that "the predicted oscillation time is greater than the experimentally observed value" and link it to the drop spreading at the bottom instead of in the middle in the axisymmetric model.
- **Interface method.** MULES (here) and PLIC (paper) treat the interface differently near the contact line, which affects whether the drop lifts off.
- **Mesh.** Period and final height change by only 2–3% between 10 and 20 cells/mm; the lift-off is the only feature that is mesh-sensitive.

6 Use of the AI tool

An AI assistant (Claude) was used for:

- reading the paper and collecting the input data for the test case;
- explaining the VOF method and the solver settings;
- extracting data points from Figure 8 of the paper for comparison with the present work;
- suggesting the interface method: MULES with interface compression (the standard method of the OpenFOAM VoF solver) instead of the geometric PLIC reconstruction used in the paper.

No AI output was used without checking. The main checks and corrections are listed below.

Table 9: AI outputs that were verified or corrected.

AI output	Verified / corrected	How and why
Figure 9 uses the inputs of Table 2, Case 4	Verified	The Fig. 9 caption says "Case 3". Checked Table 2, the Fig. 7b and Fig. 8 captions and the text of the paper: the 16.57 ms / 2.2 mm drop is water on Teflon (Case 4 in Table 2).
Time step $\Delta t \leq 4\,\mu$s	Verified	Calculated the capillary limit myself (1.2×10^{-5} s at 0.05 mm) and compared with the paper's value.
Explanation of p_{rgh}	Corrected	The AI first said p_{rgh} is smoother across the interface. From the definition $p_{rgh} = p - \rho \mathbf{g} \cdot \mathbf{h}$ it also jumps there; the real reason is hydrostatic balance and simpler boundary conditions.

AI output	Verified / corrected	How and why
Period from the peak finder	Corrected	The first version counted a small bump (and, on the coarse mesh, the wobble of the airborne drop) as a cycle. Now uses peaks after 40 ms; tested on a sine wave of known period and cross-checked with a damped-sine fit.
Final height of 2.2 mm	Checked	Calculated the height of a resting drop of the same volume at 110° (1.85 mm), which shows 2.2 mm needs a larger contact angle (used in the discussion).
Interface method: MULES instead of PLIC	Verified	Checked that MULES is the default interface method of OpenFOAM's `incompressibleVoF` solver, while the paper used PLIC in Fluent. The liquid volume was conserved to better than 0.07%, and the effect of the different method is discussed in Section 5.

7 Conclusions

- The VOF model with CSF surface tension and a contact-angle wall condition reproduces the spreading, recoil, rebound and oscillation seen in Figure 9.
- The first impact agrees well with the experiment: maximum height within about 1% and maximum spread within 5%, better than the paper's own simulation for the height.
- Like the paper's simulation, our model gives a longer oscillation period than the experiment (22.4 against 16.57 ms; paper's simulation 18.6 ms) and a lower final height (1.98 against 2.2 mm; paper's simulation about 1.8 mm).
- Results change by only 2–3% between 10 and 20 cells/mm, and the liquid volume is conserved to better than 0.1%.
- The contact angle has the largest effect. A fixed angle cannot match all measurements; a model with different advancing and receding angles would be the next step.

References

[1] P. R. Gunjal, V. V. Ranade, R. V. Chaudhari, Dynamics of drop impact on solid surface: experiments and VOF simulations, *AIChE Journal* 51(1) (2005) 59–78.

[2] J. U. Brackbill, D. B. Kothe, C. Zemach, A continuum method for modeling surface tension, *J. Comput. Phys.* 100 (1992) 335–354.

[3] C. W. Hirt, B. D. Nichols, Volume of fluid (VOF) method for the dynamics of free boundaries, *J. Comput. Phys.* 39 (1981) 201–225.

[4] The OpenFOAM Foundation, OpenFOAM 13 User Guide, `https://openfoam.org`.